

Progress Report - July 9, 2026

Built an automated two step pipeline that looks at every PhenoCam site (2023-2024) for spatial uniformity and NDVI-GCC phenology agreement, and ranks the best validation sites.

Data preparation (one time reading from phenocam api, cached data)

- Check the PhenoCam API for all site with 2023-2024 Data → 'year_check.json'
- For each site fetch lat/lon/year and compute phenophase_gap_days, dtw_per_step, divergence_score, plus the four NDVI & GCC
- phenophase dates (SOS/MOS/DOS/EOS) → site_metadata.json.
- Drop any entry with a null score → site_metadata_clean.json (end with 1,281 clean site years)

run_pipeline.py reads everything from the cached site_metadata_clean.json

Step 1: Spatial uniformity (Earth Engine)

Goal: keep only sites where the area around the camera looks the same everywhere, so the single camera represents the whole coarse satellite pixel.

- A. I want to pick the right moment for calculating uniformity. I used the stored NDVI MOS/DOS dates to center the window at peak greenness (the summer plateau, midpoint of MOS → DOS, didn't measure uniformity in winter.)
- B. Grabbing one clean image by making a ~4 km box around the camera point and the median of all Sentinel-2 scenes in a ± 30 day window with <5% cloud.
- C. Take the CV of NDVI ($\text{stdDev} / \text{mean}$) across all pixels in the box.
 - Low CV = the whole footprint greens up uniformly
- D. Checking surface type. measure the fraction of the box that is water (MNDWI > 0) or urban (NDBI > 0)
 - Modified Normalized Difference Water Index
 - Normalized Difference Built up Index
- E. Pass rule (for now). A site passes only if **CV < 0.1 AND water < 5% AND urban < 5%** (thresholds in config.py). Every site is written to step1_uniformity.json with its cv_ndvi, water_pct, urban_pct, summer_doy and summer_source, so you can see why it passed or failed.

Step 2: Phenology agreement (survivors only)

Run only on Step 1 survivors. Check if this site's NDVI and GCC describe the same season. (these are already calculated):

- phenophase_gap_days - average day gap between NDVI and GCC across the four phases. Small = they transition on the same dates.
- dtw_per_step - dynamic time warping distance between the two normalized daily curves. Small = same shape

- divergence_score - the two combined (lower = better). This is the final ranking key.

Config thresholds

- NDVI spatial CV max: 0.1
- Water fraction max: 5%
- Urban fraction max: 5%
- Sentinel-2 cloud filter: <5% · 4 km box (2 km buffer) · ±30-day summer window.

Results

- Evaluated: 1,281 site-years
- Passed Step 1 (spatially uniform): 38
- Passed in Step 2 (phenology): 38

The 38 sites below cleared both screens, ranked by divergence score (lower = stronger NDVI-GCC agreement). 34 are in the USA (marked in the USA column); the rest are international.

Key:

Green : Sites also from National Ecological Observatory Network (NEON) also part of the Ground-based Observations for Validation (GBOV) network

Golden Site: Isolated sites, no close neighbor, in USA, always safe to pick in a group if needed to show in presentations (similar to Scarlett)

#	Site (ROI)	Yr	Lat	Lon	USA	NDVI CV	Water%	Urban%	Divergence
1	alercecosteroforest_EN_1001	2023	-40.1726	-73.4439		0.062	0.0	0.0	0.171
2	congoflux_EB_1000	2023	0.8144	24.5024		0.083	0.0	2.2	0.295
3	NEON.D07.GRSM.DP1.00033_DB_1000	2023	35.6890	-83.5019	USA	0.095	0.0	0.7	0.519
4	arsltarucbec1_AG_1000	2023	40.7537	-78.0057	USA	0.100	0.0	0.9	0.535
5	harvardfarmnorth_AG_1000	2023	42.5205	-72.1822	USA	0.098	0.0	0.8	0.743
6	NEON.D07.MLBS.DP1.00042_UN_1000	2024	37.3783	-80.5248	USA	0.086	0.0	0.1	0.856
7	NEON.D07.GRSM.DP1.00042_UN_1000	2023	35.6890	-83.5019	USA	0.095	0.0	0.7	0.880
8	NEON.D01.HARV.DP1.00033_DB_1000	2024	42.5369	-72.1727	USA	0.076	0.0	0.3	0.929
9	robinson2_DB_1000	2023	37.4671	-83.1576	USA	0.055	0.0	0.1	1.036
10	NEON.D07.MLBS.DP1.00033_DB_2000	2024	37.3783	-80.5248	USA	0.086	0.0	0.1	1.054
11	harvardems2_DB_1000	2024	42.5378	-72.1715	USA	0.073	0.0	0.3	1.109
12	willowcreek_DB_1000	2023	45.8060	-90.0791	USA	0.074	0.3	0.3	1.171

[illegible]

I am assuming National Ecological Observatory Network (GBOV) is the community's "gold standard" validation sites

which should exactly be the end goal of this uniformity test + NDVI/GCC agreement test (i think?).

So I want more overlap of these sites, but right now...

Most GBOV sites fail the 4 km uniformity test (only 3/23 pass). This is interesting, It likely means either my thresholds (CV < 0.1, water/urban < 5%) are stricter than GBOV's criteria, or

These NEON tower footprints are genuinely heterogeneous at 4 km but are still valid for tower/point-scale validation (GBOV may use a smaller footprint or different products).

Should my CV cutoff be relaxed? or whether "GBOV-approved" and "spatially uniform at 4 km" are simply answering different questions.

Need more time playing with the threshold →

- NDVI spatial CV max: 0.1
- Water fraction max: 5%
- Urban fraction max: 5%

Seems like most reasons is because CV was too high, so definitely increase max CV threshold and maybe increase urban and water.

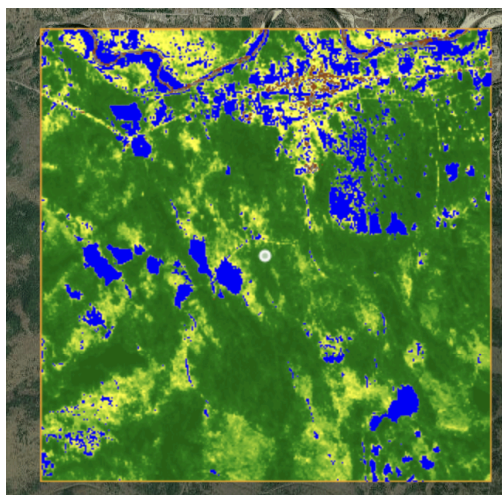
Site	Year	CV	Water%	Urban%	Failed reasons
BART_DB_1000	2023	0.135	0.2	1.2	CV too high
BART_DB_1000	2024	0.187	2.6	1.3	CV too high
HARV_DB_1000	2023	0.165	1.2	0.5	CV too high
HARV_DB_1000	2024	0.076	0.0	0.3	PASSED
KONA_AG_1000	2023	0.270	0.0	28.9	CV + urban
KONA_AG_1000	2024	0.433	0.0	54.7	CV + urban
ORNL_DB_1000	2023	0.270	0.9	8.4	CV + urban
ORNL_DB_1000	2024	0.268	0.9	6.7	CV + urban
DELA_DB_1000	2023	0.454	10.7	7.0	CV + water + urban (all three)
DELA_DB_1000	2024	0.427	10.7	6.0	CV + water + urban (all three)

TALL_EN_1000	2023	0.083	0.0	0.3	PASSED
TALL_EN_1000	2024	null	null	null	Earth Engine error, no cloud free image in the window → maybe make bigger window
CPER_GR_1000	2023	0.186	0.0	87.7	CV + urban
CPER_GR_1000	2024	0.153	0.0	100.0	CV + urban
MOAB_GR_1000	2023	0.183	0.0	99.8	CV + urban
MOAB_GR_1000	2024	0.183	0.0	99.9	CV + urban
JORN_GR_1000	2024	0.068	0.0	100.0	urban only (CV was fine)
SRER_SH_1000	2023	0.215	0.0	99.9	CV + urban
SRER_SH_1000	2024	0.189	0.0	98.9	CV + urban
ONAQ_SH_1000	2023	0.331	0.0	97.5	CV + urban
ONAQ_SH_1000	2024	0.218	0.0	99.3	CV + urban

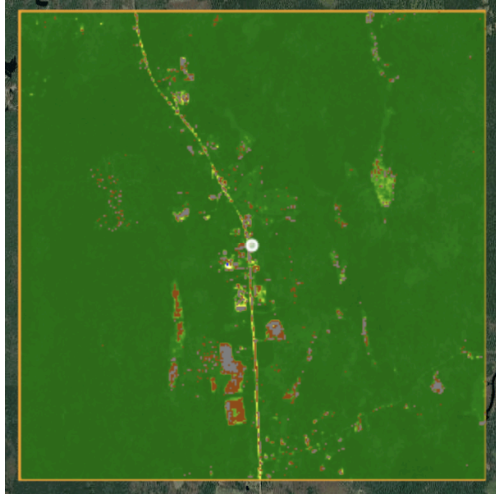
Example site that pass:

Key: Ndvi layer + urban, water, soil

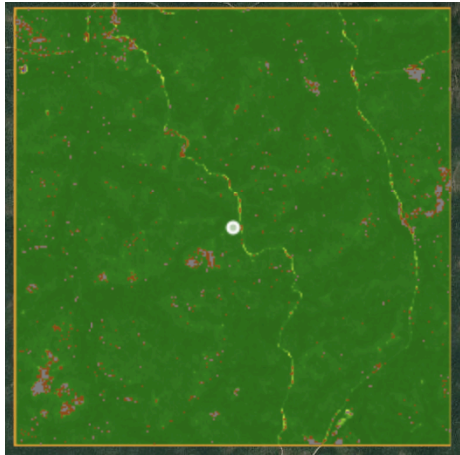
Site: alercecosteroforest_EN_1001 (2023)



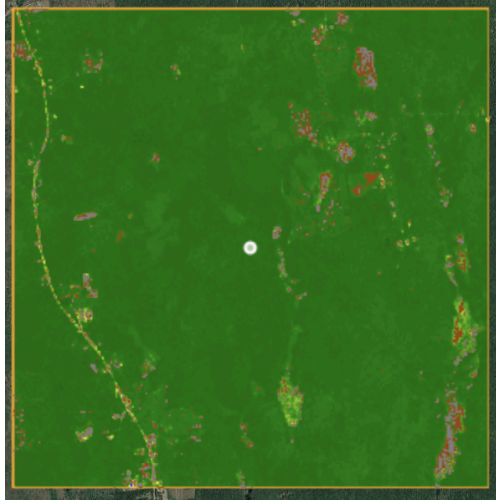
Site: harvardfarmnorth_AG_1000 (2023)



Site: NEON.D08.TALL.DP1.00033_EN_1 000 (GBOV) (2023)



Site: NEON.D01.HARV.DP1.00033_DB_ 1000 (GBOV) (2024)

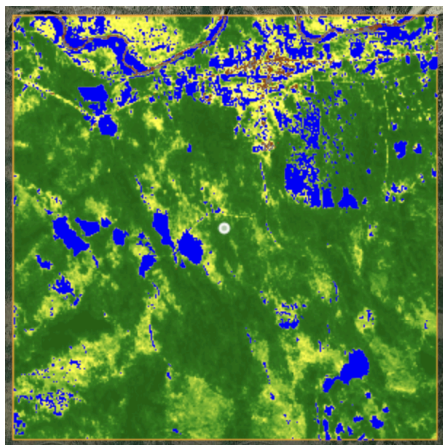


Site: tfforest_DB_1001



Example site that fail from GBOV:

Site: BART_DB_1000 (2023)



Site: DELA_DB_1000 (2023)



Site: MOAB_GR_1000 (2024)

